Contents

Chapter 1

The One-Dimensional Tight-Binding Chain

1.1 Introduction

The one-dimensional tight-binding chain is the simplest model that contains all the essential ingredients of a quantum scattering problem. Although simple, it already illustrates the complete workflow adopted throughout the `QTransport` project.

The purpose of this chapter is to show how a physical model is translated into a numerical solver. The path followed here is

$$\text{equation} \longrightarrow \text{algorithm} \longrightarrow \text{code} \longrightarrow \text{physics test.}$$

The same structure will later be used for matrix-valued Hamiltonians, graphene systems, and Bogoliubov–de Gennes problems.

1.2 Tight-Binding Hamiltonian

Consider a one-dimensional lattice with nearest-neighbor hopping. The Hamiltonian is

$$H = \sum_j \epsilon_j c_j^\dagger c_j - t \sum_j \left(c_{j+1}^\dagger c_j + c_j^\dagger c_{j+1} \right), \tag{1.1}$$

where t is the hopping energy and ϵ_j is the onsite energy at site j.

The stationary Schrödinger equation is

$$H|\psi\rangle = E|\psi\rangle\,. \tag{1.2}$$

Projecting Eq. (1.2) onto the lattice site basis gives

$$E\psi_j = \epsilon_j\psi_j - t\psi_{j-1} - t\psi_{j+1}. \tag{1.3}$$

Equivalently,

$$(E-\epsilon_j)\psi_j + t\psi_{j-1} + t\psi_{j+1} = 0. \tag{1.4}$$

Equation (1.4) is the basic equation implemented by the first version of the solver.

1.3 Infinite Periodic Chain

In the leads, the onsite energy is constant,

$$\epsilon_j = \epsilon_0. \tag{1.5}$$

For a periodic chain, Bloch solutions have the form

$$\psi_j = e^{ikj}. \tag{1.6}$$

Substituting this expression into Eq. (1.3) gives

$$Ee^{ikj} = \epsilon_0 e^{ikj} - te^{ik(j-1)} - te^{ik(j+1)}. \tag{1.7}$$

Dividing by e^{ikj},

$$E = \epsilon_0 - te^{-ik} - te^{ik}. \tag{1.8}$$

Therefore, the dispersion relation is

$$E = \epsilon_0 - 2t\cos k. \tag{1.9}$$

Solving for k gives

$$k = \arccos\left(\frac{\epsilon_0 - E}{2t}\right). \tag{1.10}$$

The corresponding group velocity is

$$v(k) = \frac{dE}{dk} = 2t\sin k. \tag{1.11}$$

In the current implementation, this logic is represented by the class `OneDimensionalLead` in `src/qtransport/leads.py`.

1.4 Scattering Geometry

The system is divided into three regions:

1. a left semi-infinite lead;
2. a finite scattering region;
3. a right semi-infinite lead.

The scattering region contains N sites,

$$j = 0, 1, \ldots, N-1. \tag{1.12}$$

The left lead occupies the sites

$$j \leq -1, \tag{1.13}$$

and the right lead occupies the sites

$$j \geq N. \tag{1.14}$$

An incoming wave from the left lead is partially reflected and partially transmitted through the scattering region.

1.5 Scattering Ansatz

For an incoming wave from the left, the wavefunction in the left lead is

$$\psi_j = e^{ikj} + re^{-ikj}, \qquad j \leq -1, \tag{1.15}$$

where r is the reflection amplitude.

In the right lead, only an outgoing transmitted wave is allowed:

$$\psi_j = \tau e^{ikj}, \qquad j \geq N, \tag{1.16}$$

where τ is the transmission amplitude.

The unknowns of the scattering problem are

$$\psi_0, \psi_1, \ldots, \psi_{N-1}, r, \tau. \tag{1.17}$$

Thus the unknown vector is

$$x = (\psi_0, \psi_1, \ldots, \psi_{N-1}, r, \tau)^T. \tag{1.18}$$

1.6 Boundary Matching

The Schrödinger equation is imposed at all sites in the scattering region and also at the two boundary sites adjacent to it.

For each site inside the scattering region,

$$(E - \epsilon_j)\psi_j + t\psi_{j-1} + t\psi_{j+1} = 0. \tag{1.19}$$

At the left edge of the scattering region, the site $j = 0$ is coupled to the lead site $j = -1$. From Eq. (1.15),

$$\psi_{-1} = e^{-ik} + re^{ik}. \tag{1.20}$$

At the right edge, the site $j = N - 1$ is coupled to the lead site $j = N$. From Eq. (1.16),

$$\psi_N = \tau e^{ikN}. \tag{1.21}$$

These substitutions convert the scattering problem into a finite linear system.

1.7 Linear System

The equations are assembled in the form

$$Ax = b, \tag{1.22}$$

where A is the coefficient matrix, x is the vector of unknown amplitudes, and b contains the contribution of the incoming wave.

The matrix has dimension

$$(N + 2) \times (N + 2). \tag{1.23}$$

The additional two equations come from imposing the tight-binding equation at the boundary sites $j = -1$ and $j = N$.

The numerical solution is obtained using

$$x = A^{-1}b, \tag{1.24}$$

but the code does not compute the inverse explicitly. Instead, it uses `numpy.linalg.solve`, which is more stable numerically.

1.8 Explicit Form of the Linear System

For clarity, let us write the structure of the matrix system explicitly. The unknown vector is

$$x = (\psi_0, \psi_1, \ldots, \psi_{N-1}, r, \tau)^T. \tag{1.25}$$

The coefficient matrix has the form

$$A = \begin{pmatrix} t & 0 & 0 & \cdots & 0 & a_L & 0 \\ E-\epsilon_0 & t & 0 & \cdots & 0 & te^{ik} & 0 \\ t & E-\epsilon_1 & t & \cdots & 0 & 0 & 0 \\ 0 & t & E-\epsilon_2 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & t & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & E-\epsilon_{N-1} & 0 & te^{ikN} \\ 0 & 0 & 0 & \cdots & t & 0 & a_R \end{pmatrix}, \tag{1.26}$$

where the last two columns correspond to r and τ, respectively. The boundary coefficients are

$$a_L = Ee^{ik} + te^{2ik}, \tag{1.27}$$

and

$$a_R = Ee^{ikN} + te^{ik(N+1)}. \tag{1.28}$$

The right-hand side contains only the incoming wave contribution:

$$b = \begin{pmatrix} -\left(Ee^{-ik} + te^{-2ik}\right) \\ -te^{-ik} \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix}. \tag{1.29}$$

This explicit form makes the role of the boundary matching clear. The bulk rows contain the tight-binding equation inside the scattering region, while the first and last rows impose the Schrödinger equation at the boundary sites of the leads.

The important point is that the incoming wave appears in the vector b, whereas the unknown reflected and transmitted amplitudes appear as columns of the matrix A.

1.9 Observables

For identical left and right leads, the reflection and transmission probabilities are

$$R = |r|^2, \tag{1.30}$$

and

$$T = |\tau|^2. \tag{1.31}$$

Probability conservation requires

$$R + T = 1. \tag{1.32}$$

For non-identical leads, the transmission probability must include the ratio of group velocities,

$$T = \frac{v_R}{v_L}|\tau|^2. \tag{1.33}$$

This generalization will become important for graphene and Bogoliubov–de Gennes systems.

1.10 Physics Tests

The first implementation is validated using physics-based tests rather than fixed numerical values.

The tests include:

1. probability conservation, $R + T = 1$;
2. absence of reflection in a homogeneous chain;
3. partial transmission through a potential barrier;
4. rejection of energies outside the propagating band.

The homogeneous-chain test is especially important. If the scattering region is identical to the leads, then translational invariance implies

$$R = 0, \qquad T = 1. \tag{1.34}$$

This test detected the first sign error in the boundary equations of the solver.

1.11 From Equation to Code

The discrete Schrödinger equation,

$$(E - \epsilon_j)\psi_j + t\psi_{j-1} + t\psi_{j+1} = 0, \tag{1.35}$$

is translated into one row of the linear system. In the implementation, the diagonal coefficient corresponds to

$$(E - \epsilon_j), \tag{1.36}$$

while the nearest-neighbor coefficients correspond to the hopping parameter t.

Thus the mathematical equation determines the structure of the matrix row assembled in the solver.

The current implementation is distributed across three files:

- `src/qtransport/leads.py`;
- `src/qtransport/solver.py`;
- `src/qtransport/observables.py`.

1.12 Software Design Notes

The first implementation introduces three basic ideas.

1. A lead is an object that knows its dispersion relation and group velocity.
2. A solver is an object that assembles and solves the boundary matching problem.
3. Observables are computed separately from the solver, so that the numerical solution and the physical interpretation remain distinct.

This separation will be preserved in later chapters. When the scalar chain is generalized to graphene or superconducting systems, the wavefunction will become a vector and the hopping constants will become matrices, but the conceptual structure will remain the same.

1.13 Lessons Learned

This first model establishes several principles that will guide the rest of the project.

- A scattering problem can be reduced to a finite linear system once the semi-infinite leads are represented analytically.
- The lead modes determine how boundary conditions are imposed.
- Physics-based tests are essential for detecting implementation errors.
- The one-dimensional chain is the minimal prototype of the general quantum transport architecture.